Patient ID	: 10124/9218	Sample ID	: 2412438
Name	**: MRS FATIMA**	Registration Date	: 10/12/2024 09:56 AM
Age/Gender	: 67 Yrs/Female	Report Date	: 10/12/2024 06:33 PM
Ref. Dr.	:	10124/9218	
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HAEMATOLOGY REPORT

Test Description	Result	Unit	Biological Reference Ranges
COMPLETE BLOOD PICTURE			
Hemoglobin	**10.7**	gm/dL	11.0 - 15.0
RBC Count	4.11	mil/cu.mm	3.8 - 5.5
Hematocrit HCT	**33.1**	%	36 - 48
Platelet Count	2.76	lac/cmm	1.5 - 4.5
Red Cell Indices			
Mean Corp Volume MCV	**80.5**	fL	83 - 101
Mean Corp Hb MCH	**26.0**	pg	27 - 32
Mean Corp Hb Conc MCHC	32.3	gm/dL	31.5 - 34.5
RDW-CV	**15.1**	%	11.6 - 14.0
Total Count and DIFFERENTIAL LEUCOCYTE COUNT			
Total WBC Count	**12800**	cell/cu.mm	4000 - 10000
Neutrophils	72	%	40 - 75
Lymphocytes	25	%	20 - 40
Eosinophils	01	%	01 - 06
Monocytes	02	%	02 - 10
Basophils	00	%	00 - 02
Microscopic Blood Picture			
RBC Morphology	**Anisocytosis Microcytic Hypochromic**		
WBC Morphology	**Leukocytosis**		
Platelet on smear	Adequate		

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Patient ID	: 10124/9218	Sample ID	: 2412438
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Ref. Dr.	:	10124/9218	
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ESR	**48**	mm/hr	0 - 22

Method : Automated Westergren

INTERPRETATION :

1. It indicates presence and intensity of an inflammatory process, never diagnostic of a specific disease. Changes are more significant than a single abnormal test.
2. It is a prognostic test and used to monitor the course or response to treatment of diseases like tuberculosis, bacterial endocarditis, acute rheumatic fever, rheumatoid arthritis, SLE, Hodgkins disease, temporal arteritis, polymyalgia rheumatica.
3. It is also increased in pregnancy, multiple myeloma, menstruation, and hypothyroidism.

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Patient ID	: 10124/9218	Sample ID	: 2412438
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Ref. Dr.	:	10124/9218	
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HBA1C/GLYCOCYLATED

HbA1c Glycosilated Haemoglobin	**7.4**	%	4 - 6

Method: HPLC, NGSP certified

Estimated Average Glucose : 166 mg/dL

As per American Diabetes Association (ADA)

Reference Group	HbA1c in %
Non diabetic adults >=18 years	<5.7
At risk (Prediabetes)	5.7 - 6.4
Diagnosing Diabetes	>= 6.5
Therapeutic goals for glycemic control	Age > 19 years Goal of therapy: < 7.0 Action suggested: > 8.0 Age < 19 years Goal of therapy: <7.5

ADA criteria for correlation

HbA1c(%)	Mean Plasma Glucose (mg/dL)
6	126
7	154
8	183
9	212
10	240
11	269
12	298

Note:

1. Since HbA1c reflects long term fluctuations in the blood glucose concentration, a diabetic patient who is recently under good control may still have a high concentration of HbA1c. Converse is true for a diabetic previously under good control but now poorly controlled .
2. Target goals of < 7.0 % may be beneficial in patients with short duration of diabetes, long life expectancy and no significant cardiovascular disease. In patients with significant complications of diabetes, limited life expectancy or extensive co-morbid conditions, targeting a goal of < 7.0 % may not be appropriate.

Comments: HbA1c provides an index of average blood glucose levels over the past 8 - 12 weeks and is a much better indicator of long term glycemic control as compared to blood and urinary glucose determinations.

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Patient ID	: 10124/9218	Sample ID	: 2412438
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Ref. Dr.	:	10124/9218	
Center	: Q LABS DIAGNOSTICS		

BIOCHEMISTRY REPORT

Test Description	Result	Unit	Biological Reference Ranges
KIDNEY FUNCTION TEST(KFT)			
Serum Creatinine *Method: Modified Jaffe with no deproteinzation*	0.80	mg/dL	0.60 - 1.40
Urea *Method: Urease UV GLDH*	21.64	mg/dl	10.0 - 45.0
Blood Urea Nitrogen-BUN *Method: Calculated*	8.7	mg/dl	7 - 20
Uric Acid *Method: Uricase Peroxidase*	4.34	mg/dl	2.5 - 6.8
Sodium *Method: ISE*	141	mEq/L	135 - 146
Potassium *Method: ISE*	4.0	mEq/L	3.5 - 5.5
Chloride	103	mEq/L	96.0 - 109

Interpretation :

Kidney function tests help to screen the individual for renal disease and to determine the extent or progression of renal disease. These tests also aid in determining drug dosage for drugs excreted through the kidneys. The clinical syndrome resulting from decreased renal function and azotemia is called uremia Renal azotemia: glomerular nephritis and chronic pyelonephritis. Prerenal azotemia: severe dehydration, hemorrhagic shock, and excessive protein intake. Post renal azotemia: urethral stones or tumors and prostatic obstructions Measurement of urea in dialysis fluids is widely used in assessing the adequacy of renal replacement therapy.

In these prerenal situations, the plasma creatinine concentration may be normal.In obstructive post renal conditions, both plasma creatinine and urea concentrations will be increased, although there is often a greater increase in plasma urea than creatinine because of the increased back diffusion.These considerations give rise to the principal clinical utility of plasma urea, which lies in its measurement in conjunction with that of plasma creatinine and subsequent calculation of the urea nitrogen/creatinine ratio. This ratio has been used as a crude discriminator between prerenal and postrenal azotemia. Significantly lower ratios usually denote (1) acute tubular necrosis, (2) low protein intake, (3) starvation, or (4) severe liver disease (decreased urea synthesis). So even though blood urea is not an excellent marker of renal dysfunction as it rises quite late in the dysfunction and its rise is also not exclusive to kidney dysfunction, but for practical purposes serum urea level is still one of the most ordered test and forms an important part of the kidney function test.

Concentrations in excess of 6.0 mg/dL at 32 weeks gestation have been noted to be associated with a high perinatal mortality rate.

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Authorised Signature

Patient ID	: 10124/9218	Sample ID	: 2412438
Name	**: MRS FATIMA**	Registration Date	: 10/12/2024 09:56 AM
Age/Gender	: 67 Yrs/Female	Report Date	: 10/12/2024 06:33 PM
Ref. Dr.	:	10124/9218	
Center	: Q LABS DIAGNOSTICS		

Fasting Plasma Glucose	109	mg/dl	Normal : 74 - 106
Method: Hexokinase			Impaired GT : 107 - 125
			Diabetic : >or=126

> or =126	Diabetic	Its a group of metabolic disorders characterized by a high blood sugar level over a prolonged period of time
101-125	Impaired GT	Impaired glucose tolerance means that blood glucose is raised beyond normal levels, but not high enough to warrant a diabetes diagnosis. With impaired glucose tolerance one can face a much greater risk of developing diabetes and cardiovascular disease.
70-100	Normal	The American Diabetes Association recommends a routine screening for type 2 diabetes. If the results are normal, the screening should be repeated every 3 years.

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Patient ID	: 10124/9218	Sample ID	: 2412438
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Ref. Dr.	:	10124/9218	
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LIPID PROFILE

Cholesterol-Total	108.86	mg/dL	< 200 - Desirable
Method: Spectrophotometry			200 - 239 -Boderline High
			> 240 - High
Triglycerides level	111.56	mg/dL	< 150 - Normal
Method: Serum, Enzymatic, endpoint			150 - 199 -Boderline High
			200 - 499 -High
			>500 Very -High
HDL Cholesterol	21.8	mg/dL	< 40 - Low
Method: Serum, Direct measure-PEG			> 40 - Normal
LDL Cholesterol	64.75	mg/dL	< 100 - Optimal
Method: Enzymatic selective protection			100 - 129 - Near/Above Optimal
			130 - 159 - Borderline high
			160 - 189 - High
			> 190 - Very High
VLDL Cholesterol	22.31	mg/dL	6 - 38
Method: Serum, Enzymatic			
CHOL/HDL RATIO	4.99		0.27 - 5.0
Method: Serum, Enzymatic			
LDL/HDL RATIO	2.97		2.5 - 3.5
Method: Serum, Enzymatic			

NOTE

8-10 hours fasting sample is required

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Age/Gender	: 67 Yrs/Female	Report Date	: 10/12/2024 06:33 PM
Ref. Dr.	:	10124/9218	
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LIVER FUNCTION TEST (LFT)

TOTAL BILIRUBIN	1.04	mg/dl	0.1 -1.2
Method: Serum, Jendrassik Grof			
DIRECT BILIRUBIN	0.30	mg/dL	0.0 -0.3
Method: Serum, Diazotization			
INDIRECT BILIRUBIN	0.74	mg/dl	0.0 -1.0
Method: Serum, Calculated			
SGPT (ALT)	13.41	U/L	0 - 49
Method: Serum, UV with P5P, IFCC 37 degree			
SGOT (AST)	16.55	U/L	0 - 46
Method: Serum, UV with P5P, IFCC 37 degree			
ALKALINE PHOSPHATASE	59.13	U/L	42 - 125
Method: DGKC			
TOTAL PROTEIN	6.74	g/dl	6.0 - 8.3
Method: Serum, Biuret, reagent blank end point			
SERUM ALBUMIN	3.88	g/dl	3.4 - 5.4
Method: Serum, Bromocresol green			
SERUM GLOBULIN	2.86	g/dl	1.8 - 3.6
Method: Serum, Calculated			
A/G RATIO	1.36		1.2 - 2.2
Method: Serum, Calculated			

NOTE :

In known cases of Chronic Liver disease due to Viral Hepatitis B & C, Alcoholic liver disease or Non alcoholic fatty liver disease, Enhanced liver fibrosis (ELF) test may be used to evaluate liver fibrosis.

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Authorised Signature

Patient ID	: 10124/9218	Sample ID	: 2412438
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Age/Gender	: 67 Yrs/Female	Report Date	: 10/12/2024 06:33 PM
Ref. Dr.	:	10124/9218	
Center	: Q LABS DIAGNOSTICS		

Total Calcium- Serum	8.76	mg/dl	8.4 - 10.5

Method : AresenazoIII

Hypocalcemia:

Long-term therapy must be tailored to the specific disease causing the hypocalcemia. The therapeutic endpoint is to achieve a serum calcium level of 8.0 to 8.5 mg/dL to prevent tetany. For symptomatic hypocalcemia, calcium may be administered intravenously

Hypercalcemia:

The level at which hypercalcemic symptoms occur varies from patient to patient. Symptoms are common when serum calcium levels are above 11.5 mg/dL, although patients may be asymptomatic at this level. Levels above 12.0 mg/dL are considered a critical value in the Mayo Health System. Severe hypercalcemia (>15.0 mg/dL) is a medical emergency.

IRON STUDIES

SERUM IRON	**34.8**	ugm/dl	40 - 160
TIBC	**192**	ugm/dl	220 - 440
UIBC	157	ugm/dl	
Transferrin saturation	**18**	%	20 - 50

Method : Biochemical

Test Usage:

To aid in evaluation of a number of conditions involving red cell production and destruction, iron metabolism, or iron transport.

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Patient ID	: 10124/9218	Sample ID	: 2412438
Name	**: MRS FATIMA**	Registration Date	: 10/12/2024 09:56 AM
Age/Gender	: 67 Yrs/Female	Report Date	: 10/12/2024 06:33 PM
Ref. Dr.	:	10124/9218	
Center	: Q LABS DIAGNOSTICS		

COAGULATION STUDY REPORT

Test Description	Result	Unit	Biological Reference Ranges
PROTHROMBIN TIME (PTINR)			
Prothrombin Time for Patient's Plasma	**23.28**	Sec.	10 - 17
Prothrombin Time for control plasma	13.5	Sec.	
Prothombin Index	57.99	%	
International Normalised Ratio INR	**1.82**		0.8 - 1.5
ISI value of the Prothrombin used	1.1		
Prothombin Ratio	1.82		

Interpretation :

The prothrombin time (PT) and international normalized ratio (INR) are measures of the extrinsic pathway of coagulation.The INR is used only for patients on stable oral anticoagulant therapy. It makes no significant contribution to the diagnosis or treatment of patients whose PT is prolonged for other reasons.

Increased PT times may be due to:

Factor deficiencies(X , II , V , I), Coumadin (warfarin) therapy, Liver Diseases (Bile duct obstruction, Cirrhosis , Hepatitis), Hemmorhagic Disaease of the newborn, DIC, Malabsorption, Fibrinolysis, Vitamin K deficiency.

Interference in PT/INR Alcohol,antibiotics, aspirin, cimetidine,thrombin Inhibitors(Increase PT) Barbiturates, oral contraceptives, Hormone replacement therapy (HRT), and vitamin K (Decrease PT).

Dr.SYEDA SALEHA FAREED MD PATHOLOGY
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Age/Gender	: 67 Yrs/Female	Report Date	: 10/12/2024 06:33 PM
Ref. Dr.	:	10124/9218	
Center	: Q LABS DIAGNOSTICS		

IMMUNOASSAY REPORT

Test Description	Result	Unit	Biological Reference Ranges
Vitamin B12- Serum	**1211**	pg/ml	197.0 - 771.0

Method : ECLIA

INTERPRETATION

1. Vit B12 levels are decreased in megaloblastic anemia, partial/total gastrectomy, pernicious anemia, peripheral neuropathies,chronic alcoholism, senile dementia, and treated epilepsy.
2. An associated increase in homocysteine levels is an independent risk marker for cardiovascular disease and deep vein thrombosis.
3. HoloTranscobalamin II levels are a more accurate marker of active VitB12 component.

Vitamin D- Total (25-OH-Vit D)- Serum	26.5	ng/ml	2 - 10 Defficiency
			10 - 30 Insufficiency
			30 - 100 Sufficiency
			More than - 100 Toxicity

Method : ECLIA

INTERPRETATION:

Vitamin D is a fat soluble vitamin & exists in two main forms as cholecalciferol (Vitamin D3) which is synthesized in skin from 7-dehydrocholesterol in response to sunlight exposure & Ergocalciferol (Vitamin D2) present mainly in dietary sources. Both cholecalciferol & Ergocalciferol are converted to 25 (OH) Vitamin D in liver.

Testing for 25 (OH) Vitamin D is recommended as it is the best indicator of Vitamin D nutritional status as obtained from sunlight exposure & dietary intake.

1. 25 OH Vitamin D is the best indicator of Vitamin D nutritional status, it is used as an aid in assessment of Vitamin D sufficiency in adults.
2. 25 OH Vitamin D deficiency is seen in secondary hyperparathyroidism.
3. Decreased levels of 25 OH Vitamin D can lead to Osteomalacia, reduced bone mass & thus increase the risk of bone fractures.
4. Decreased 25 OH Vitamin D levels are also associated with low bone mineral density & also seen in nutritional rickets.
5. Decreased levels of 25 OH Vitamin D are also associated with increased cardiovascular risk, low immunity & chronic renal failure.
6. Elevated levels are associated with Vitamin D intoxication.

Dr.SYEDA SALEHA FAREED MD PATHOLOGY
Authorised Signature

Patient ID	: 10124/9218	Sample ID	: 2412438
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Ref. Dr.	:	10124/9218	
Center	: Q LABS DIAGNOSTICS		

Thyroid Function Test (TFT)

T3	0.81	ng/mL	0.40 - 1.81
T4	8.15	µg/dL	5.00 - 12.60
TSH(Serum)	3.22	µIU/mL	0.50 - 5.50

Method : ECLIA

INTERPRETATION

TSH	T3 / FT3	T4 / FT4	Suggested Interpretation for the Thyroid Function Tests Pattern
Within Range	Decreased	Within Range	• Isolated Low T3-often seen in elderly & associated Non-Thyroidal illness. In elderly the drop in T3 level can be upto 25%.
Raised	Within Range	Within Range	•Isolated High TSHespecially in the range of 4.7 to 15 mIU/ml is commonly associated with Physiological & Biological TSH Variability. •Subclinical Autoimmune Hypothyroidism •Intermittent T4 therapy for hypothyroidism •Recovery phase after Non-Thyroidal illness"
Raised	Decreased	Decreased	•Chronic Autoimmune Thyroiditis •Post thyroidectomy,Post radioiodine •Hypothyroid phase of transient thyroiditis"
Raised or within Range	Raised	Raised or within Range	•Interfering antibodies to thyroid hormones (anti-TPO antibodies) •Intermittent T4 therapy or T4 overdose •Drug interference- Amiodarone, Heparin,Beta blockers,steroids,anti-epileptics"
Decreased	Raised or within Range	Raised or within Range	•Isolated Low TSH -especially in the range of 0.1 to 0.4 often seen in elderly & associated with Non-Thyroidal illness •Subclinical Hyperthyroidism •Thyroxine ingestion"
Decreased	Decreased	Decreased	•Central Hypothyroidism •Non-Thyroidal illness •Recent treatment for Hyperthyroidism (TSH remains suppressed)"
Decreased	Raised	Raised	•Primary Hyperthyroidism (Graves' disease),Multinodular goitre, Toxic nodule •Transient thyroiditis:Postpartum, Silent (lymphocytic), Postviral (granulomatous,subacute, DeQuervain's),Gestational thyrotoxicosis with hyperemesis gravidarum"
Decreased or within Range	Raised	Within Range	•T3 toxicosis •Non-Thyroidal illness

References: 1. Interpretation of thyroid function tests. Dayan et al. THE LANCET • Vol 357 • February 24, 2001
2. Laboratory Evaluation of Thyroid Function, Indian Thyroid Guidelines, JAPI, January 2011,vol. 59

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Ref. Dr.	:	10124/9218	
Center	: Q LABS DIAGNOSTICS		

URINE EXAMINATION REPORT

Test Description	Result	Unit	Biological Reference Ranges
URINE ROUTINE			
Physical Examination			
Colour	Pale Yellow		Pale Yellow
Apperance	Slightly Turbid		Clear
Specific Gravity	1.015		
Chemical Examination			
Blood	Absent		Absent
Sugar	NIL		Absent
Protein	Trace		Absent
Ketone Bodies	Absent		Absent
Bile Salt	Absent		Absent
Bile Pigment	Absent		Absent
Reaction(pH)	6.0		4.5 - 8.0
Microscopic Examination			
Bacteria	Not seen		Not Seen
Epithelial Cells	4-6/hpf	/hpf	1-2/hpf
Pus cells	2-3/hpf	/hpf	2-3/hpf
RBC's	NIL	/hpf	Nil
Casts	Not Seen		Not Seen
Crystals	Absent		Absent
Yeast Cells	Not seen		Not seen

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Ref. Dr.	:	10124/9218	
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Other Finding -

**** End of the report. ****
Please correlate clinically.

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